

```
# =====
# CONCEPT MAP: SAMPLING, ALIASING & IMAGE DISTORTION
# Google Colab Code
# =====
# Add text cell

# Install Graphviz
!apt-get -qq install graphviz
!pip -q install graphviz

from graphviz import Digraph
from IPython.display import Image, display

# -----
# Create Concept Map
# -----
dot = Digraph("SamplingAliasing", format="png")

# Graph settings
dot.attr(rankdir="TB")
dot.attr(size="16,16")
dot.attr(dpi="300")
dot.attr(nodesep="0.5")
dot.attr(ranksep="0.8")
dot.attr(splines="ortho")

# =====
# MAIN NODE
# =====
```

```

dot.node(
    "A",
    "SAMPLING,\nALIASING &\nIMAGE DISTORTION",
    shape="box",
    style="filled",
    fillcolor="lightblue"
    fontsize="22" Add text cell
)

# =====
# SAMPLING
# =====

dot.node(
    "B",
    "Sampling",
    shape="box",
    style="filled",
    fillcolor="lightgreen"
)

dot.node("C","Sampling Rate",shape="ellipse")
dot.node("D","Nyquist Sampling\nCriterion",shape="ellipse")
dot.node("E","Spatial Resolution",shape="ellipse")

# =====
# ALIASING
# =====

dot.node(
    "F",

```

```

    "Aliasing",
    shape="box",
    style="filled",
    fillcolor="lightcoral"
)

dot.node("G", "Un Add text cell ", shape="ellipse")
dot.node("H", "False Patterns", shape="ellipse")
dot.node("I", "Jagged Edges", shape="ellipse")
dot.node("J", "Moiré Patterns", shape="ellipse")

# =====
# IMAGE DISTORTION
# =====

dot.node(
    "K",
    "Image Distortion",
    shape="box",
    style="filled",
    fillcolor="salmon"
)

dot.node("L", "Loss of Detail", shape="ellipse")
dot.node("M", "Blurred Appearance", shape="ellipse")
dot.node("N", "Reduced Image Quality", shape="ellipse")

# =====
# SOLUTIONS
# =====

```

```

dot.node(
    "O",
    "Prevention Methods",
    shape="box",
    style="filled",
    fillcolor="khaki"
)

dot.node("P","Increase Sampling Rate",shape="ellipse")
dot.node("Q","Anti-Aliasing Filter",shape="ellipse")
dot.node("R","Higher Resolution Sensor",shape="ellipse")

# =====
# APPLICATIONS
# =====

dot.node(
    "S",
    "Applications",
    shape="box",
    style="filled",
    fillcolor="lightskyblue"
)

dot.node("T","Medical Imaging",shape="box")
dot.node("U","Remote Sensing",shape="box")
dot.node("V","Autonomous Vehicles",shape="box")
dot.node("W","Face Recognition",shape="box")
dot.node("X","Industrial Inspection",shape="box")

# =====
# LITERATURE

```

```

# LITERATURE
# =====

dot.node(
    "Y",
    "Theoretical Insights\n\n"
    "• Gonzalez Add text cell .gital Image Processing\n\n"
    "• Richard Szeliski\nComputer Vision\n\n"
    "• Forsyth & Ponce\nComputer Vision\n\n"
    "• Shannon-Nyquist Sampling Theory",
    shape="note",
    style="filled",
    fillcolor="lightyellow"
)

# =====
# CONNECTIONS
# =====

# Main Branches
dot.edge("A","B")
dot.edge("A","F")
dot.edge("A","K")

# Sampling
dot.edge("B","C")
dot.edge("C","D",label="must satisfy")
dot.edge("D","E",label="ensures")

# Cause of aliasing
dot.edge("C","G",label="low rate causes")
dot.edge("G","F")

```

```
# Effects of aliasing
dot.edge("F","H")
dot.edge("F","I")
dot.edge("F","J")

# Image distortion
dot.edge("H","K")
dot.edge("I","K")
dot.edge("J","K")

dot.edge("K","L")
dot.edge("K","M")
dot.edge("K","N")

# Prevention
dot.edge("N","O",label="minimized by")
dot.edge("O","P")
dot.edge("O","Q")
dot.edge("O","R")

# Applications
dot.edge("O","S",label="improves")
dot.edge("S","T")
dot.edge("S","U")
dot.edge("S","V")
dot.edge("S","W")
dot.edge("S","X")

# Literature
dot.edge("Y","A",label="supported by")
```

```
# =====  
# SAVE & DISPLAY  
# =====  
  
filename = dot.r Add text cell ling_Aliasing_Image_Distortion_Concept_Map")  
  
display(Image(filename))  
  
print("Concept Map generated successfully!")
```

Add text cell

Warning: Orthogonal edges do not currently handle edge labels. Try using xlabel.

